SIT313 – Secure Frontend Applications

Task C1: New Post Page – Project Report

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1. Introduction

The purpose of Task C1 was to extend the DEV@Deakin web platform by implementing a dynamic New Post page using React. This task focused on developing interactive user interfaces through React States, React Hooks, and conditional rendering.

The New Post page allows users to select between two post types: Question and Article. Based on the selected option, the interface dynamically displays the corresponding input fields.

In addition to the New Post functionality, I improved the overall website design by maintaining reusable components, adding navigation elements and improving the footer.

2. Project Objectives

The main objectives of this task were:

- To create a New Post page using React.
- To implement Question and Article selection.
- To use React States and Hooks for dynamic interaction.
- To implement conditional rendering based on the selected post type.
- To validate user input before accepting a submission.
- To display appropriate success and error messages.
- To maintain a responsive and user-friendly interface.

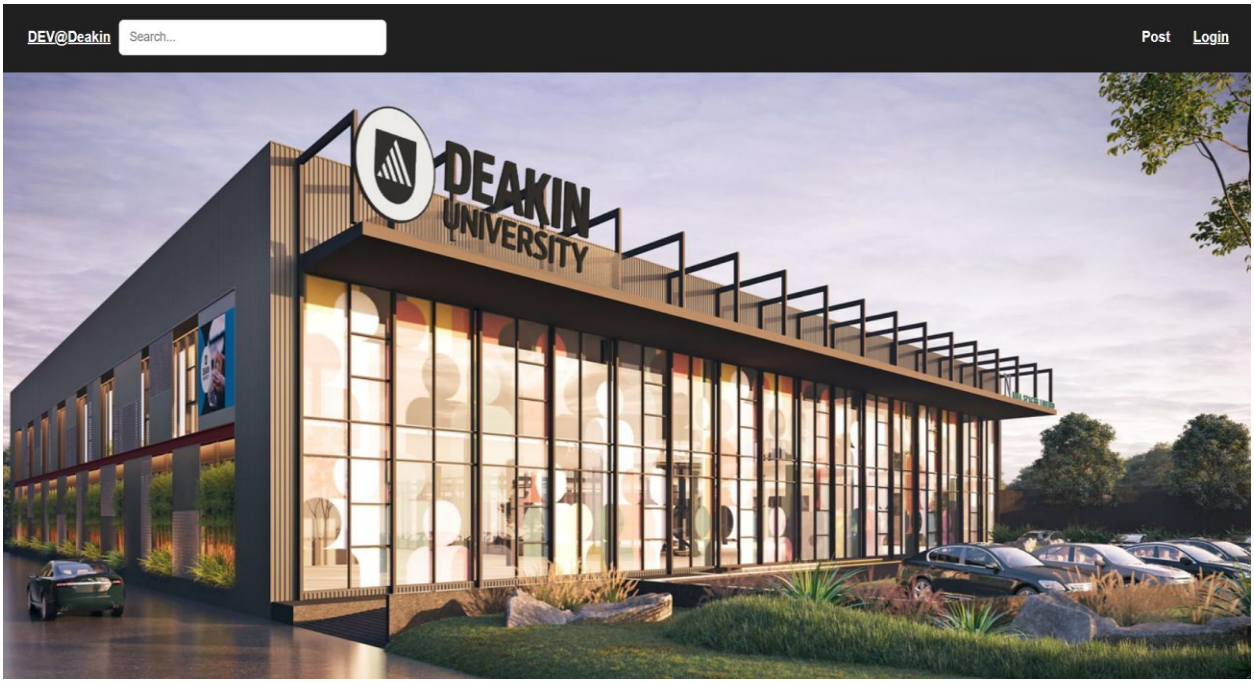
3. Website Homepage

The DEV@Deakin homepage provides a structured interface containing a navigation bar, banner, featured articles, featured tutorials, and footer.

The website uses a consistent colour scheme and reusable React components to create a clear and organised layout.

The navigation bar provides access to the main sections of the website, including the New Post page.

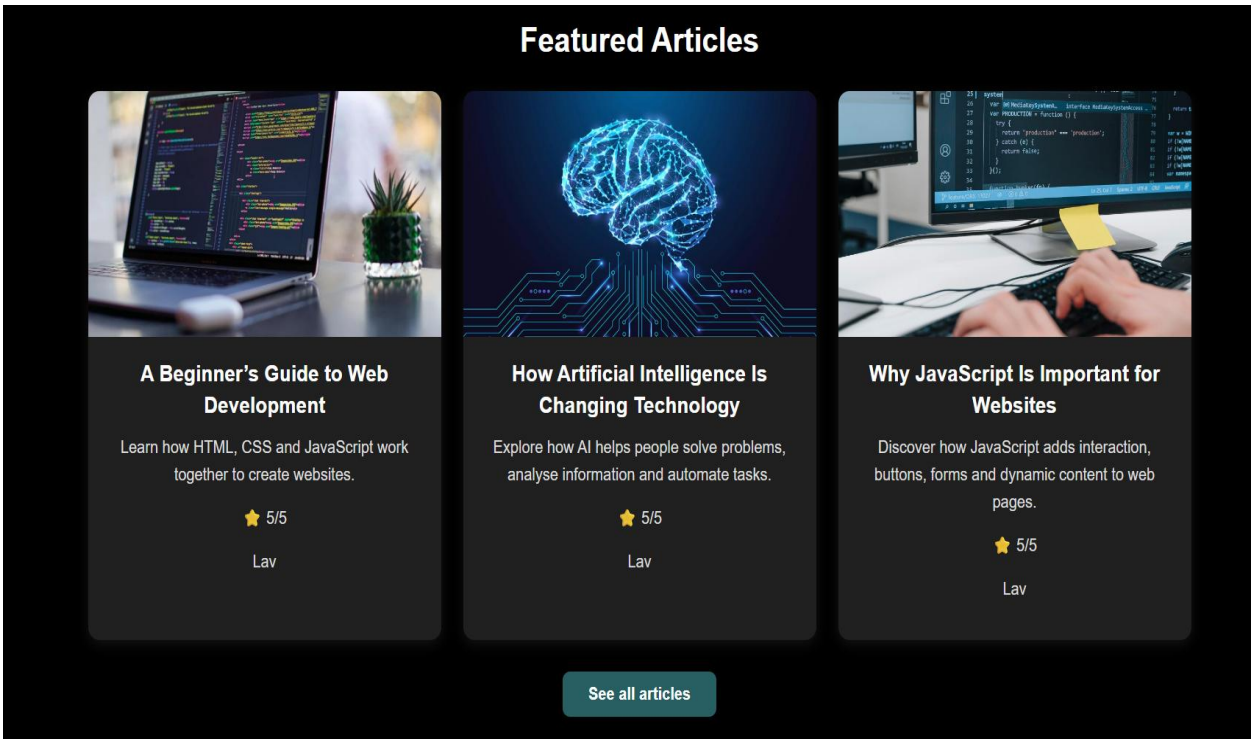
Screenshot:



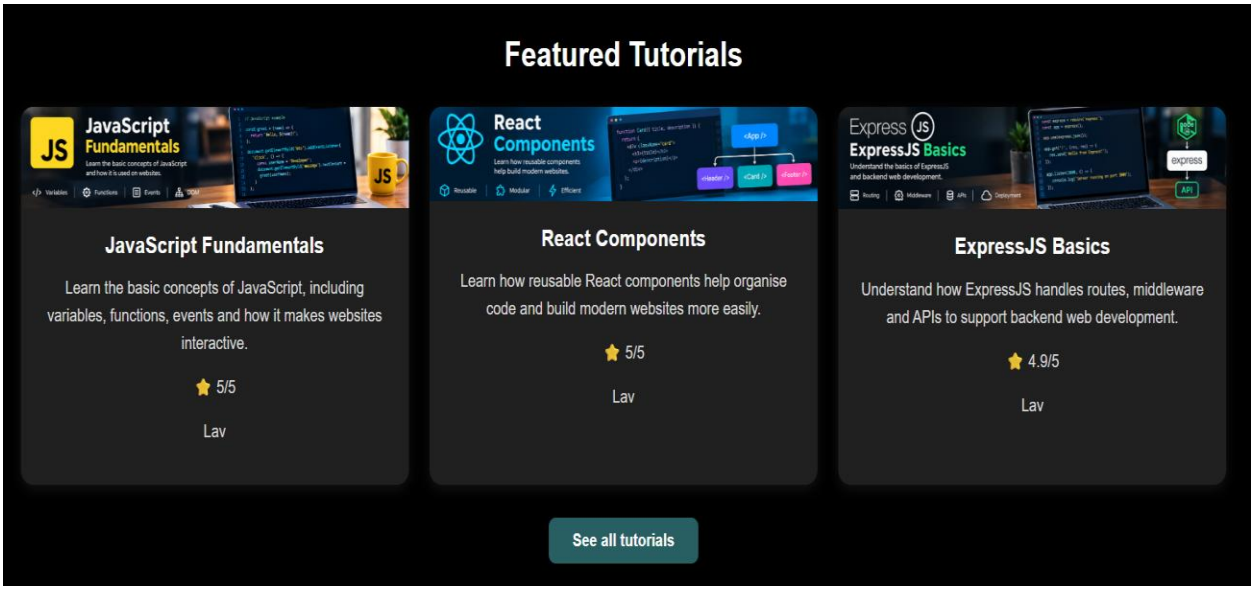
4. Featured Articles and Tutorials

Following the feedback received, I also added the See all articles and See all tutorials buttons beneath their respective card lists.

Screenshot: Articles



Screenshot: Featured Tutorials



5. New Post Page Implementation

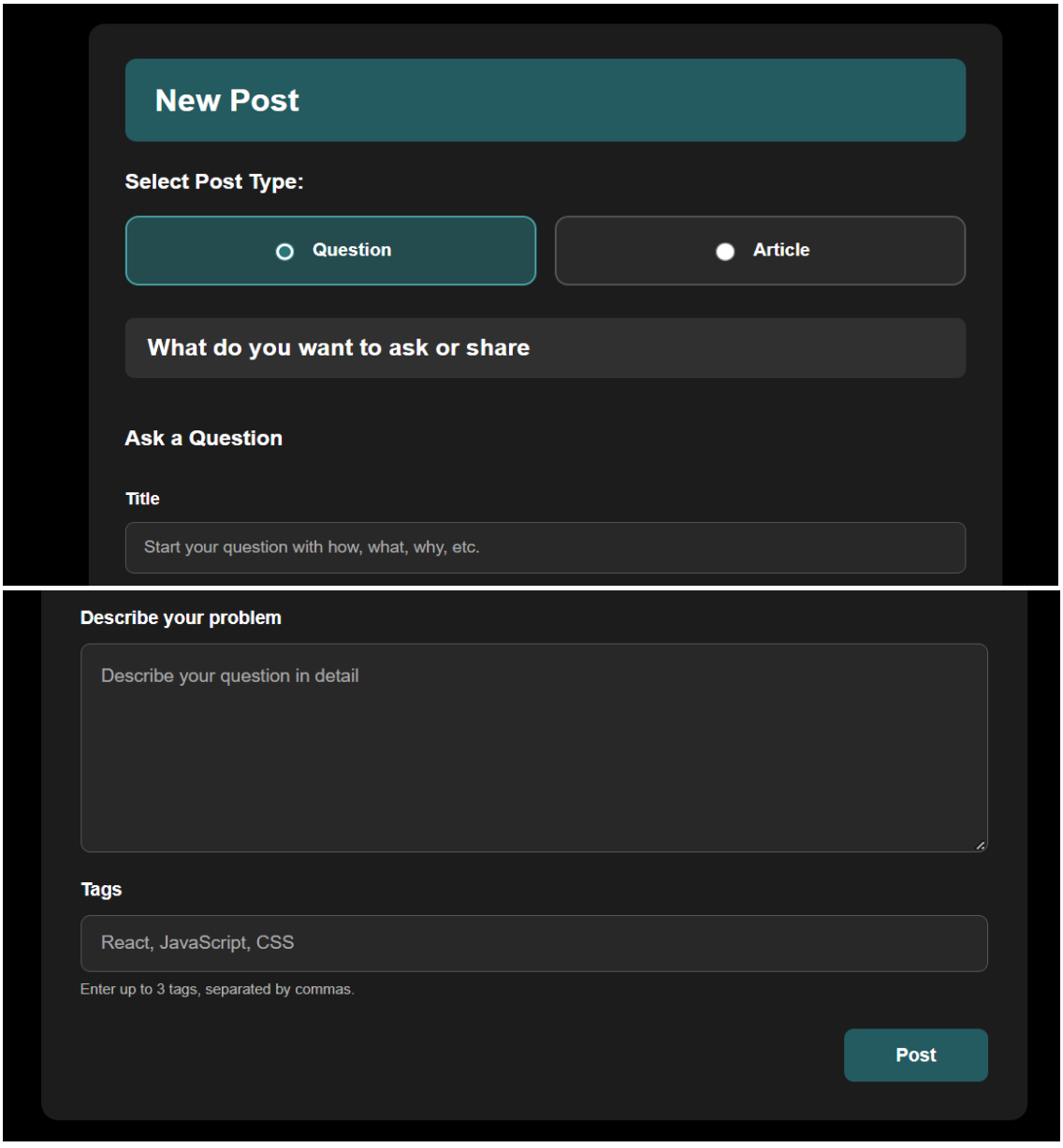
It provides users with two options for creating content:

- **Question:** Allows users to prepare a question using the relevant input fields.
- **Article:** Allows users to prepare an article using the relevant input fields.

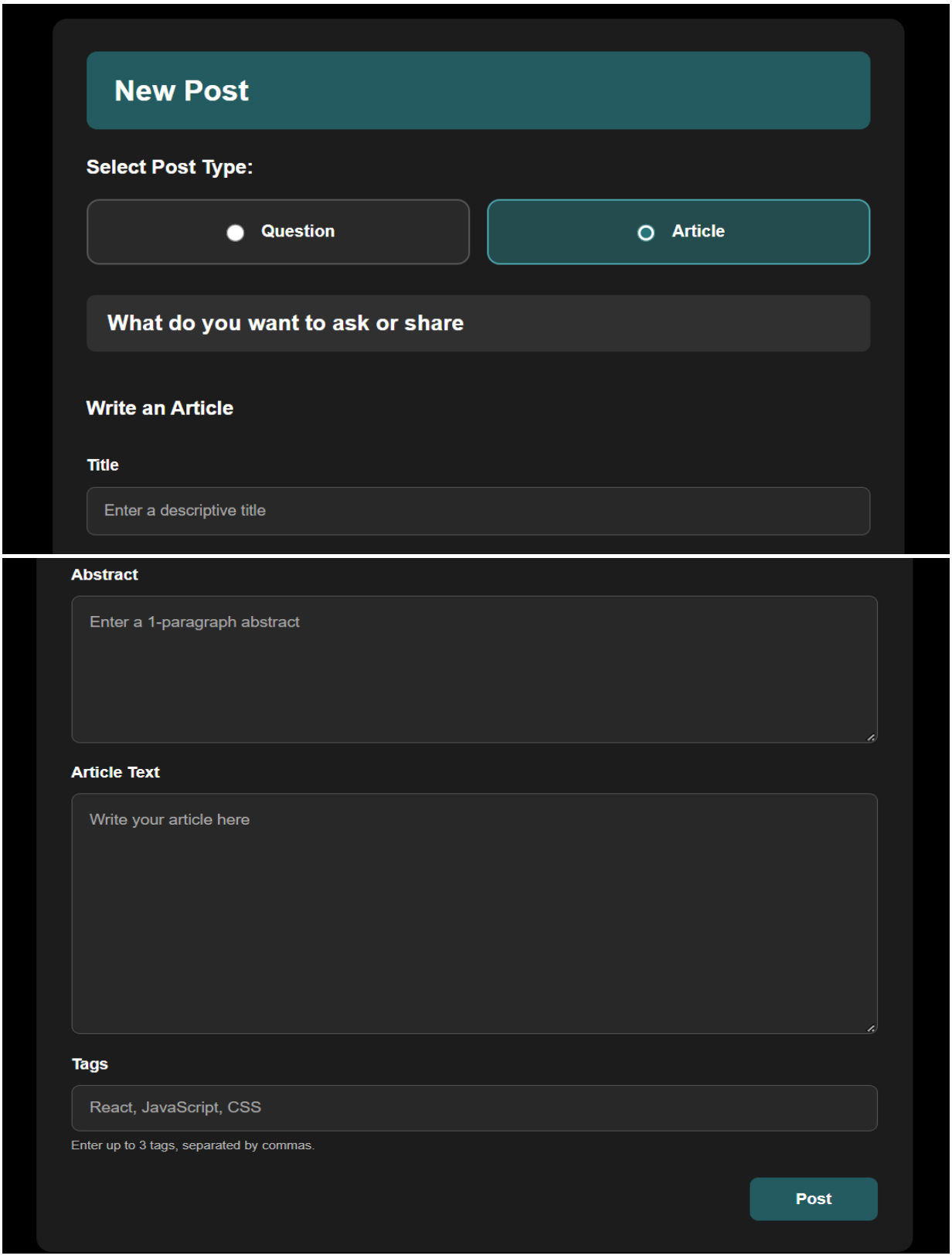
The page changes dynamically when the user selects a different post type.

This functionality was implemented using React States and conditional rendering.

Screenshot: Question Form



Screenshot: Article Form



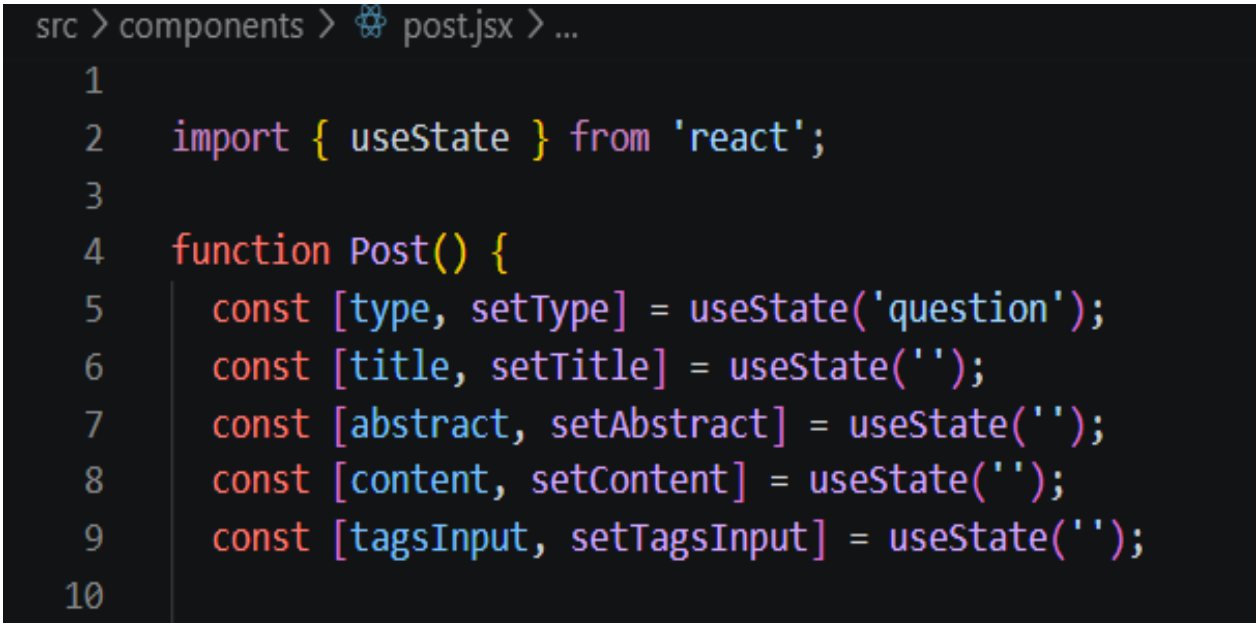
6. React States and Conditional Rendering

React States and Hooks were used to control the behaviour of the New Post page.

The selected post type is maintained in the component state. When the user changes the selection, React updates the displayed form without requiring a complete page refresh.

Conditional rendering ensures that the appropriate interface is displayed for each post type.

Screenshot: React useState Hooks



Screenshot: Conditional Rendering

```
<h3>
  {type === 'question'
    ? 'Ask a Question'
    : 'Write an Article'}
</h3>
```

```
{type === 'article' && (
  <>
    <label htmlFor="article-abstract">
      Abstract
    </label>

    <textarea
      id="article-abstract"
      rows={3}
      value={abstract}
      onChange={(event) => setAbstract(event.target.value)}
      placeholder="Enter a 1-paragraph abstract"
    />
  </>
)}
```

7. Form Validation

Form validation was implemented to improve the reliability and usability of the New Post page.

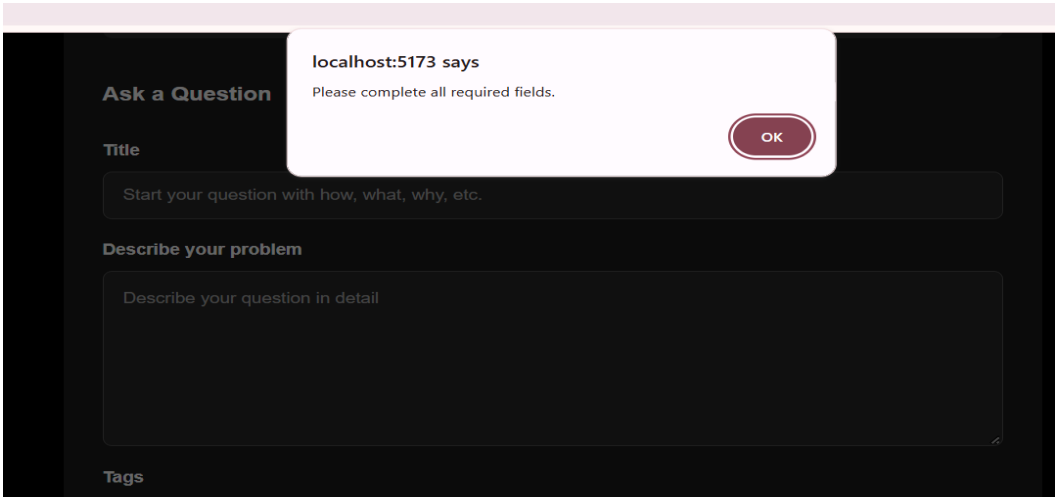
Before a submission is considered successful, the entered information is checked to ensure that the required fields contain valid data.

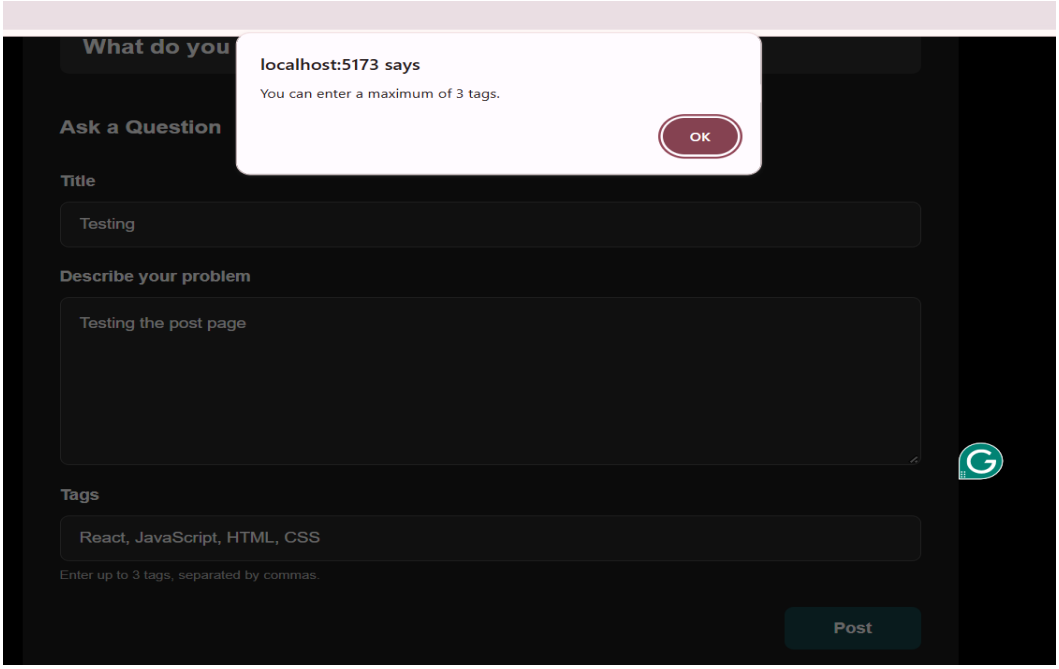
The form also includes validation for tags, with a maximum of three tags.

If the user submits incomplete or invalid information, an error message is displayed.

This prevents invalid submissions and helps users understand which information needs to be corrected.

Screenshot:





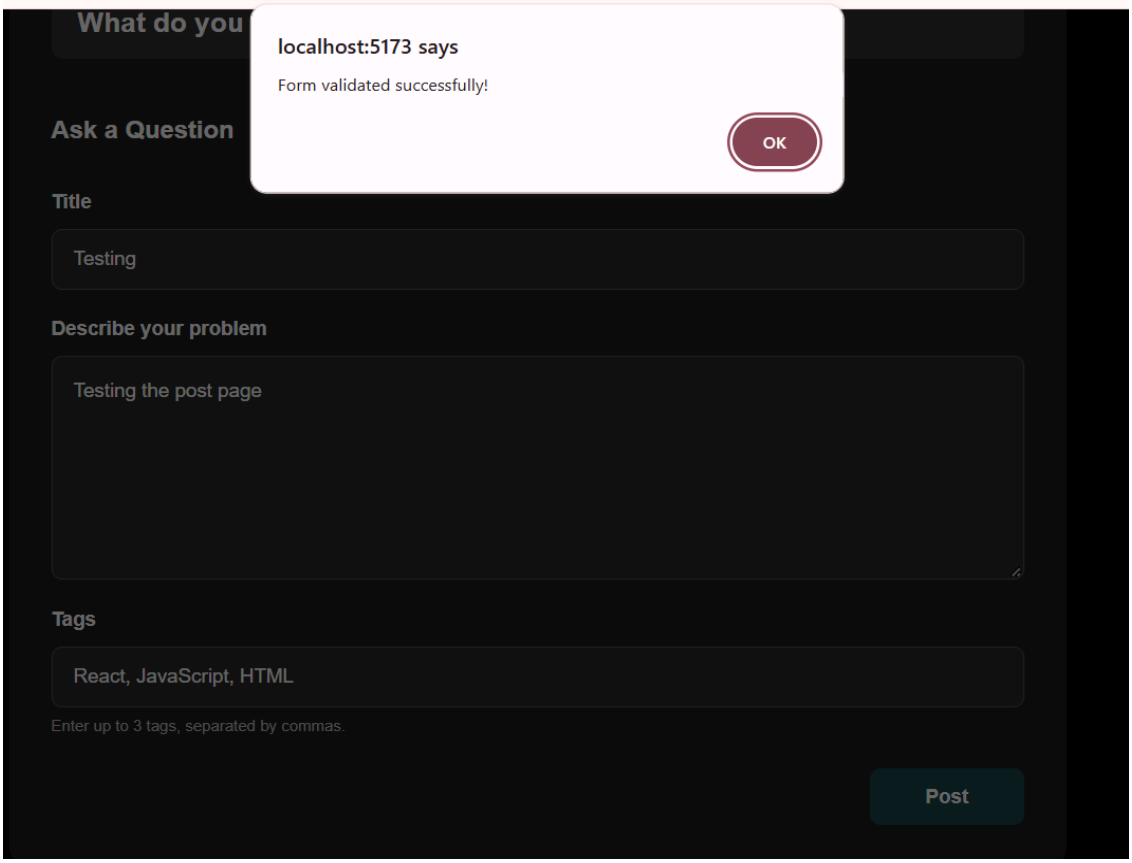
8. Post Submission Feedback

The New Post page includes a Post button that provides feedback after the user attempts to submit the form.

If the information passes validation, the application displays a success alert. If validation fails, the user receives an error message.

The task focuses on frontend functionality, so the Post button demonstrates the submission process without storing posts in a database or publishing them permanently.

Screenshot:



9. Footer Design and Social Media Icons

The footer was designed using three main columns:

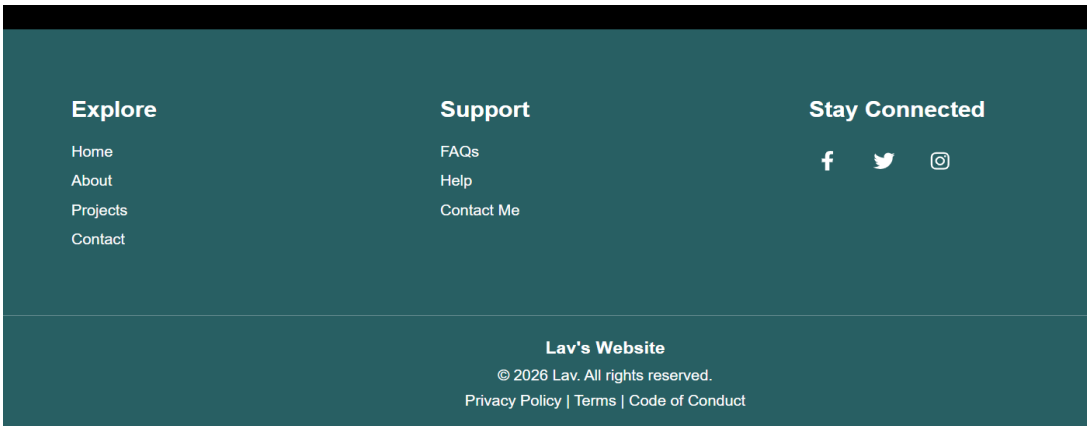
- **Explore:** Provides navigation links such as Home, About, Projects, and Contact.
- **Support:** Contains FAQs, Help, and Contact Me links.
- **Stay Connected:** Displays social media links.

Based on the feedback received, the original text-based social media links were replaced with icons for Facebook, Twitter, and Instagram.

The footer layout was also adjusted using CSS to improve the alignment and spacing between the three columns.

These changes improve the visual presentation and consistency of the website.

Screenshot:



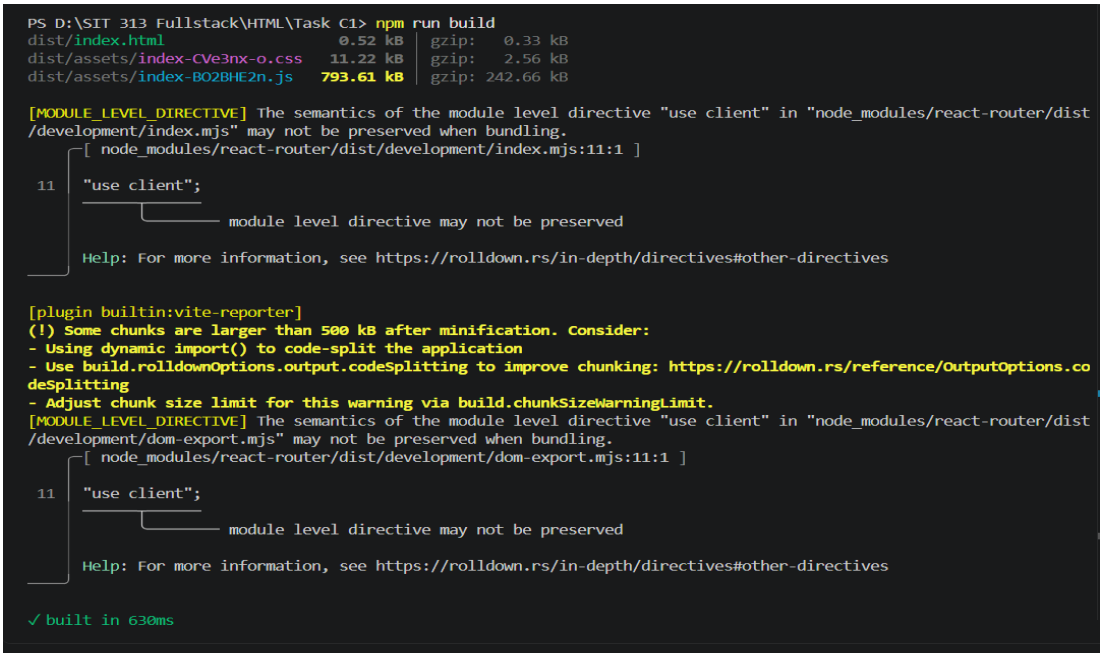
10. Testing and Verification

The website was tested using the Vite development server to check its functionality and appearance. The testing process included verifying the Question and Article selection, checking conditional rendering, validating required fields, and confirming that success and error messages were displayed correctly.

The Featured Articles, Featured Tutorials, and footer sections were also reviewed to confirm that the updated elements appeared correctly.

The project was additionally checked using the npm run build command to verify that Vite could generate a production build.

Screenshot:



11. GitHub Repository

Git was used to manage the source code and track changes during development.

The project was uploaded to GitHub, allowing the React components, CSS files, and project configuration to be stored in a central repository.

A .gitignore file was included to prevent unnecessary files, such as the node_modules/ directory, from being committed to the repository.

GitHub Repository Link:

https://github.com/lavsukhija001-sudo/SIT313---Task-C1

13. Conclusion

Task C1 helped me develop a better understanding of interactive frontend development using React.

By implementing the New Post page, I gained practical experience with React States, Hooks, conditional rendering, and form validation.

I also improved my understanding of reusable components, CSS styling, responsive layouts.

The feedback received during development helped me improve the Featured Articles and Tutorials sections and the footer design.

Overall, this task provided valuable experience in developing a structured, interactive, and maintainable React web application.